LaCONIC: A Label-Aware and Graph-Guided Multi-Omics Collaborative Learning Model for Cancer Survival Prediction –Supplementary Materials

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A Dataset and preprocessing

Expression data preprocessing (performed before data splitting). To construct a high-quality and prognosis-relevant dataset, we performed all preprocessing *prior* to train/validation/test partitioning. For each cancer cohort, we applied a stringent QC and feature screening pipeline: (1) removing genes with missing values in more than 10% of samples; (2) excluding samples with more than 50% missing measurements; (3) filtering low-expression genes by retaining those with expression > 1 in at least 20% of samples; (4) applying $\log_2(x+1)$ transformation; and (5) conducting univariable Cox regression on Overall Survival (OS) to select prognosis-associated genes (e.g., $p < 0.05$). After data splitting, we performed per-gene z-score standardization using statistics computed on the training set only, and applied the same transformation to the validation and test sets to prevent information leakage. This procedure yielded a stable, denoised, and survival-informative feature space, ensuring consistent downstream training and fair comparisons across models.

miRNA synergy quantification (Target overlap–PPI–functional consistency). We quantified miRNA-pair synergy following the main line of *shared targets–protein interactions–functional consistency*. First, we computed a target similarity score (TSS) based on the overlap of shared target genes, and derive a protein interaction score (PIS) by measuring the interaction strength between the encoded proteins of the target genes in a PPI network. Second, we assigned *synergistic* versus *non-synergistic* labels to each miRNA pair using Gene Ontology Biological Process (GOBP) enrichment results, providing a function-level and externally verifiable supervision signal. Third, we integrated TSS and PIS using a weighted logistic regression, where the coefficients are learned via five-fold cross-validation, to compute a synergy score for each miRNA pair by jointly accounting for regulatory target similarity and topological interaction evidence. Finally, we applied min–max normalization to the synergy scores and retain pairs with scores greater than 0.5 to construct a high-confidence synergy network. The resulting miRNA–miRNA network preserved biologically meaningful miRNA-mediated regulatory semantics while exhibiting a topology closer to functional synergy patterns, thereby providing a more reliable network prior for subsequent LaCONIC survival modeling.

B Topological Feature Initialization

The initial node-level topological features is constructed from the betweenness centrality of the ceRNA network. Betweenness centrality is defined as

$$C_B(v) = \frac{1}{(G-1)(G-2)} \sum_{\substack{s,t\in\mathcal{V}\\ s\neq t\neq v}} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{1}$$

where σ_{st} denotes the number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths that pass through node v. This metric reflects the 'bridge' role of a node in signal transduction and regulatory pathways, and thus serves as an important topological feature for identifying key regulatory genes and hub miRNAs.

Given the high computational cost of exact betweenness computation in large-scale biological networks, we adopt an adaptive sampling approximation based on Brandes' algorithm [1]. Specifically, the number of sampled source nodes k is determined dynamically according to the number of nodes G and network density ρ:

$$k = \begin{cases} \min\big(50\log_{10}(G{+}1)\times 2,\ k_{\max}\big), & \rho < 0.01, \\ \max\big(50\log_{10}(G{+}1)/2,\ k_{\min}\big), & \rho \geq 0.01. \end{cases} \tag{2}$$

We then randomly sample k source nodes from the graph and use breadth-first search to compute node dependencies, which are accumulated and averaged to obtain the approximate betweenness centrality:

$$\tilde{C}_B(v) = \frac{1}{k(G-1)(G-2)} \sum_{s=1}^{k} \delta_v^{(s)}, \tag{3}$$

where $\delta_v^{(s)}$ denotes the dependency of node v with respect to the s-th sampled source node. Finally, the resulting values are assembled into the topological feature matrix $\boldsymbol{b} = \tilde{C}_B(v) \in \mathbb{R}^g$, which is fed into the GCIL module to enhance the network awareness of node representations.

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C Experimental setup

C.1 Metrics

To systematically evaluate the model's performance in cancer survival prediction and risk stratification tasks, this study used various clinical survival analysis and statistical testing metrics, including Harrell's C-Index, iAUC, IPCW C-Index, and risk stratification, to comprehensively characterize the model's predictive ability and stability across different levels.

Harrell's Concordance Index (Harrell's C-Index). The Harrell's C-Index is used to assess the consistency of the model's ranking of individual survival times [4], defined as follows:

$$C_{\text{Harrell}} = \frac{1}{N_{\text{comp}}} \sum_{i,j} \mathbb{I}\big[(t_i < t_j) \wedge (r_i > r_j)\big], \tag{4}$$

where t_i is the survival time of sample i; r_i is the model-predicted risk score; and N_{comp} is the number of comparable sample pairs, which only include pairs of samples with an event (death), excluding censored samples.

Integrated AUC (iAUC). The iAUC is obtained by integrating the time-dependent AUC over time, used to evaluate the model's discriminatory ability at different follow-up time points [5], defined as:

$$\text{iAUC} = \frac{1}{T} \int_0^T \text{AUC}(t)\, dt, \tag{5}$$

where $\text{AUC}(t)$ represents the instantaneous predictive performance at time t, calculated based on cumulative dynamic AUC. The time evaluation points are selected as the 25%, 50%, and 75% percentiles of the survival time distribution.

Inverse Probability of Censoring Weighting C-Index (IPCW C-Index). The IPCW C-Index is used to reduce the bias caused by censored data [20], defined as:

$$C_{\text{IPCW}} = \sum_{i<j} w_{ij}\, \mathbb{I}\big[(t_i < t_j) \wedge (\hat{r}_i > \hat{r}_j)\big], \tag{6}$$

where w_{ij} is the censoring adjustment weight based on individual survival probability estimates.

Kaplan–Meier (KM) Risk Stratification Analysis. KM risk stratification divides samples into high-risk and low-risk groups based on the model's predicted median risk, and KM survival curves are plotted to show survival differences between the groups [8, 16]. The significance of group differences is analyzed using the Log-rank test [14], and the p-value is used to assess statistical differences between the survival curves of high- and low-risk groups. To quantify the mortality risk of the high-risk group relative to the low-risk group, the Hazard Ratio (HR) and its 95% confidence interval (CI) are calculated using the Cox proportional hazards model. HR and its confidence interval are derived when both groups experience events, reflecting the model's discriminatory ability and stability in survival risk stratification.

C.2 Dataset division and sampling

To ensure the representativeness and stability of multi-view cancer data during the training and validation phases, this study adopted a strategy combining stratified partitioning and balanced batch sampling to mitigate issues arising from event censoring and imbalanced subtype distribution in survival analysis. Data division was based on the combination of survival event status and subtype labels, ensuring a balanced distribution of categories and event ratios across subsets. The original data was first split into an 80% training set and a 20% test set, with 20% of the training set further reserved as a validation set. The final data partitioning ratio was 64% for training, 16% for validation, and 20% for testing.

During the training phase, to address the issue of event sample scarcity, we designed a balanced batch sampler to ensure that each batch contains at least k event samples, stabilizing gradient estimates. Let the batch size be B, the number of event samples be N_1, and the total number of training samples be N. The minimum number of event samples was calculated as $k = \left\lceil \frac{N_1}{N/B} \right\rceil$. This mechanism ensured an adequate proportion of event samples in each batch, enhancing the stability of Cox loss and time-dependent AUC. With this stratified partitioning and balanced batch strategy, the model's robustness in sample distribution and risk estimation was significantly improved.

C.3 Training strategy of LaCONIC

To achieve stable convergence and task collaboration in the multi-task multi-omics model, LaCONIC employed a strategy combining stage-wise weight ramp-up and smooth learning rate scheduling during training. Specifically, the training was divided into three stages: In Stage 0, the model optimized only survival prediction and classification tasks to stabilize the foundational feature space; in Stage 1, the cross-task loss $\mathcal{L}_{\text{cross}}$ and supervised contrastive loss $\mathcal{L}_{\text{sup}}$ were gradually introduced, with their weights increasing linearly to align risk and subtype features collaboratively; in Stage 2, the prototype contrastive constraint $\mathcal{L}_{\text{proto}}$ was enabled, further enhancing the boundary clarity between different subtypes. This progressive scheduling mechanism effectively prevented gradient conflicts during the early stages of multi-task training, enabling the model to learn high-level feature associations while converging stably.

In addition, the AdamW optimizer was used with learning rate warmup and cosine annealing strategies to ensure smooth parameter updates and stable learning rate changes. The model's performance was evaluated after each iteration using the validation set's Harrell's C-Index and a composite loss function, with the best weights saved when performance improves. Early stopping was triggered if no improvement was observed over several consecutive iterations (default is 50) to prevent overfitting. This training strategy enhanced both task collaboration and generalization, providing an efficient and reliable optimization paradigm for cancer multi-omics survival prediction.

C.4 Baselines

To systematically evaluate the survival prediction performance of LaCONIC, we selected 14 representative survival models as competing baselines, covering both traditional statistical survival methods and deep survival models. The deep methods are further categorized into discrete-time and continuous-time paradigms according to their time modeling schemes.

Traditional survival methods include Coxnet, CoxBoost, and RSF. Coxnet extends the CoxPH model with ℓ_1/ℓ_2/Elastic Net regularization, enabling robust estimation and variable selection in high-dimensional settings [18]. CoxBoost augments the Cox model under a boosting framework by iteratively updating the predictor, improving flexibility for complex risk structures and high-dimensional data [23]. RSF (Random Survival Forest) generalizes random forests to survival analysis and captures nonlinearities and interactions via ensemble trees while naturally handling censoring [7].

Deep discrete-time models include DeepHit, Transformer Survival, Trans-STG, and SurvTRACE. DeepHit discretizes survival time into intervals and learns the survival distribution, together with a ranking-consistency term to enhance risk ordering [11]. Transformer Survival leverages Transformer architectures to model nonlinear interactions in high-dimensional omics features and outputs discrete-time survival probabilities [6]. Trans-STG further adopts a semi-supervised/self-training strategy that exploits unlabeled samples to strengthen survival representation learning and generalization [19]. SurvTRACE is a Transformer-based model tailored for survival analysis, emphasizing stable risk representation and time modeling, and has demonstrated strong robustness across multi-cancer tasks [21].

Deep continuous-time models include DeepSurv, Cox-nnet, DCAP, HFBSurv, CAMR, FGCNSurv, and PCLSurv. DeepSurv parameterizes the Cox risk function with a deep neural network to enable nonlinear risk modeling in continuous time [9]. Cox-nnet learns Cox-type risk representations via neural networks and offers stronger nonlinear fitting capacity for high-dimensional inputs [3]. DCAP jointly optimizes multi-omics representation learning and survival modeling to improve cross-omics fusion quality [2]. HFBSurv and CAMR enhance risk representation and interpretability through multi-omics fusion and cross-modal interaction/attention mechanisms, respectively [12, 24]. FGCNSurv builds and fuses omics-specific patient kNN graphs, then applies graph convolutional network to learn patient representations for survival prediction. [22]. PCLSurv adopts prototype contrastive learning to align semantic prototypes, thereby improving the discriminability and generalization of risk representations [13].

To ensure reproducibility and fair comparisons, we followed the recommended implementations and default settings from the original papers whenever possible. Specifically, the three traditional methods were implemented using `scikit-survival` [17]; DeepSurv and DeepHit were implemented via `PyCox` [10]; all other deep models were implemented in `PyTorch` [15]. Among them, Transformer Survival, Trans-STG, and DCAP were re-implemented in this paper following the original descriptions, while SurvTRACE, HFBSurv, CAMR, FGCNSurv, and PCLSurv were executed using their publicly available GitHub implementations.

Regarding input features, to ensure fair comparisons, all methods except FGCNSurv and PCLSurv used the same three-omics inputs (miRNA, mRNA, and lncRNA), while these two models followed their original settings and used only miRNA and mRNA. Moreover, except for HFBSurv, CAMR, FGCNSurv, and PCLSurv, the remaining models do not employ omics-specific encoders. Therefore, we concatenated the three omics features into a single matrix as input. In contrast, the above four methods encoded each omics modality separately and then performed fusion according to their original architectures.

C.5 Hyperparameter settings

To ensure a fair and efficient hyperparameter search, we employed Optuna to tune the hyperparameters of LaCONIC and all baseline models, with the random seed uniformly fixed at 42 to facilitate reproducibility. Considering that the dimensionality and search space of hyperparameters differ across models, we assigned an appropriate number of trials to each model (Detailed setting in Table 1).

Table 1: Number of Optuna hyperparameter search trials for each method

Model	trial number
LaCONIC	200
Coxnet	50
RSF	150
CoxBoost	200
DeepSurv	200
DeepHit	200
Cox-nnet	150
Transformer Survival	200
DCAP	100
SurvTRACE	200
TransSTG	200
CAMR	75
HFBSurv	75
FGCNSurv	75
PCLSurv	20

The hyperparameters of LaCONIC hyperparameters includes batch size, attention/classifier/survival dropout rates, EMA momentum, training epochs, hidden layer dimensions and depth of classifier/survival predictor, intermediate dimension in GCIL module, learning rate, number of attention heads and output dimensions in ACRL module, temperatures for cross/prototype/supervised contrastive losses, and the corresponding loss weights (cross, prototype, supervised, and classfication). Under the miRNA+mRNA+lncRNA multi-omics combination, the optimal hyperparameter settings of LaCONIC across all cancer types were summarized in Table 2, whereas the retuned optimal hyperparameters under different omics combinations for the CESC dataset were presented in Table 3. The optimal hyperparameter settings of baselines were summarized in Tables 4–Table 17.

Table 2: Best hyperparameters of LaCONIC across cancers

param	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_c_index	0.828248031	0.894339623	0.707492795	0.884422111	0.821782178	0.760298661	0.889316988
batch_size	16	16	24	16	24	24	8
dropout_att1	0.068555596	0.068563974	0.080922912	0.044009127	0.035735726	0.031921386	0.004273933
dropout_att2	0.027318465	0.045832206	0.028476266	0.027819807	0.008836704	0.020991902	0.001986249
dropout_cls	0.039415717	0.276581389	0.218186753	0.19035103	0.183115986	0.387294427	0.19984428
dropout_surv	0.135118334	0.033940272	0.041874682	0.146065789	0.060142412	0.280037287	0.080410038
ema_m	0.929585169	0.967138415	0.980374947	0.946690413	0.901331893	0.88883725	0.96526094
epochs	200	200	200	200	200	200	200
hidden_dims	[256, 224]	[128, 96]	[128]	[128]	[224, 160]	[160, 160]	[64, 64]
hidden_n_layers	2	2	1	1	2	2	2
inter_model	16	64	32	16	64	32	16
lr	0.001159917	0.002241198	0.003971425	0.000998748	0.000239019	0.000740467	0.000216812
num_heads	2	1	2	1	2	2	1
o_dims	[16, 16, 16]	[32, 32, 32]	[16, 16, 16]	[16, 16, 16]	[16, 16, 16]	[16, 16, 16]	[32, 32, 32]
temp_cross	0.126960286	0.109216443	0.084026541	0.074625476	0.073747873	0.137328299	0.101770808
temp_proto	0.113330171	0.071744504	0.062913029	0.105313326	0.102274767	0.081560676	0.066410544
temp_sup	0.195456452	0.19755224	0.152274852	0.194525396	0.12830436	0.165778486	0.206739436
weight_cross	0.160332632	0.109301884	0.055345172	0.086849955	0.178947764	0.179448773	0.170426181
weight_decay	5.95E-05	6.19E-05	1.47E-07	3.21E-07	1.78E-07	8.70E-07	1.27E-05
weight_proto	0.006468953	0.091604206	0.087769499	0.017355726	0.094696382	0.117411948	0.071058932
weight_sup	0.182471357	0.240996404	0.19505594	0.232218319	0.187838378	0.204075648	0.181349014
weight_task	0.5	0.7	0.7	0.7	0.1	0.5	0.1

Table 3: Best hyperparameters of LaCONIC under various omic combinations on CESC data

Hyperparameters	miRNA	mRNA	lncRNA	miRNA+mRNA	miRNA+lncRNA	mRNA+lncRNA	miRNA+mRNA+lncRNA
best_c_index	0.81509434	0.879245283	0.713207547	0.830188679	0.78490566	0.849056604	0.894339623
batch_size	16	16	16	16	16	16	16
dropout_att1	0.037397771	0.093294115	0.023296206	0.017024219	0.060256102	0.028225835	0.068563974
dropout_att2	0.064042786	0.091092466	0.043392725	0.000869357	0.02952621	0.008097519	0.045832206
dropout_cls	0.245826332	0.16202801	0.270829515	0.04161505	0.08635746	0.016094939	0.276581389
dropout_surv	0.027155327	0.086184121	0.195424328	0.204044094	0.06522064	0.254364128	0.033940272
ema_m	0.91025237	0.989904877	0.981549092	0.922221804	0.978033327	0.924601457	0.967138415
epochs	200	200	200	200	200	200	200
hidden_dims	[160, 96]	[160]	[192]	[224]	[160, 128]	[224]	[128, 96]
hidden_n_layers	2	1	1	1	2	1	2
inter_model	64	64	32	32	16	16	64
lr	0.001941776	0.003172307	0.00355262	0.001000495	0.004636073	0.000655382	0.002241198
num_heads	2	2	1	1	2	1	1
o_dims	[32, 32, 32]	[32, 32, 32]	[16, 16, 16]	[16, 16, 16]	[32, 32, 32]	[16, 16, 16]	[32, 32, 32]
temp_cross	0.122987302	0.075557684	0.092093164	0.092477109	0.071703794	0.102675691	0.109216443
temp_proto	0.103485087	0.090577273	0.101900742	0.100800204	0.113718338	0.118849466	0.071744504
temp_sup	0.217726647	0.200140365	0.125938892	0.155536319	0.155839416	0.166732911	0.19755224
weight_cross	0.170408792	0.100058607	0.054380153	0.129261241	0.136925993	0.051473682	0.109301884
weight_decay	1.25E-05	3.16E-07	2.14E-05	3.56E-06	5.24E-06	2.24E-07	6.19E-05
weight_proto	0.00235547	0.059342691	0.033624805	0.005207007	0.056940275	0.054306061	0.091604206
weight_sup	0.279779056	0.302003685	0.329361449	0.155685673	0.303497662	0.293145995	0.240996404
weight_task	0.9	0.3	0.3	1	1	0.3	0.7

Table 4: Best hyperparameters of CAMR across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	8.041339e-01	0.764151	6.001441e-01	8.178392e-01	7.186469e-01	0.701081	0.842032
lr	1.809691e-04	0.000834	2.170219e-03	3.083498e-03	4.169858e-03	0.000122	0.000259
lambda_reg	6.783867e-05	0.000001	1.525521e-05	1.484840e-04	5.359932e-07	0.000008	0.000001
weight_decay	3.187451e-07	0.000006	1.817557e-07	2.065507e-07	1.042297e-07	0.000012	0.000011

Table 5: Best hyperparameters of FGCNSurv across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.808961	0.826415	6.037464e-01	8.161209e-01	0.689256	6.975484e-01	0.865520
k	8.000000	8.000000	7.000000e+00	4.000000e+00	10.000000	1.000000e+01	4.000000
lr	0.000306	0.000433	4.770725e-04	9.717156e-04	0.000448	1.540745e-03	0.000842
weight_decay	0.000027	0.000071	1.456504e-07	3.167275e-07	0.000004	1.111996e-07	0.000004

Table 6: Best hyperparameters of HFBSurv across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	7.554134e-01	0.807547	0.647695	8.115578e-01	0.686469	0.669156	0.832574
lr	7.475993e-04	0.000180	0.003675	2.696155e-04	0.000543	0.000245	0.000101
lambda_reg	7.669581e-07	0.000101	0.000028	4.911117e-05	0.000001	0.000177	0.000083
weight_decay	5.987475e-06	0.000049	0.000002	7.921335e-07	0.000014	0.000027	0.000002

Table 7: Best hyperparameters of PCLSurv across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.821763	0.743396	0.644092	8.136020e-01	0.740496	0.725935	8.683287e-01
lr	0.000103	0.000769	0.000329	7.281696e-04	0.001752	0.000557	2.479390e-04
weight_decay	0.000001	0.000001	0.000004	1.376365e-07	0.000006	0.000001	2.406031e-07

Table 8: Best hyperparameters of CoxBoost across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.754921	0.741509	0.614553	0.738693	0.735974	0.599125	0.838529
learning_rate	0.042043	0.04906	0.02924	0.213336	0.273679	0.27512	0.108908
max_depth	7	6	6	3	6	3	3
max_features	sqrt	sqrt	sqrt	sqrt	sqrt	sqrt	sqrt
min_samples_leaf	7	3	8	21	19	9	21
min_samples_split	5	8	18	8	16	6	14
n_estimators	80	98	240	164	74	223	78
subsample	0.703887	0.821635	0.645642	0.769202	0.996368	0.802458	0.754964

Table 9: Best hyperparameters of Coxnet across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.812008	0.769811	0.611671	0.811558	0.691419	0.524202	0.645534
l1_ratio	0.995983	0.311345	0.214332	0.001399	0.007990	0.433012	0.987511

Table 10: Best hyperparameters of Cox-nnet across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	7.401575e-01	0.686792	0.481988	0.788945	0.676568	0.660402	0.758669
batch_size	1.600000e+01	16.000000	24.000000	16.000000	24.000000	24.000000	8.000000
dropout_rate	2.862788e-01	0.215288	0.174491	0.294369	0.122659	0.145464	0.089400
epochs	2.000000e+02	200.000000	200.000000	200.000000	200.000000	200.000000	200.000000
lr	5.686883e-04	0.004989	0.004947	0.004902	0.000580	0.001815	0.004645
weight_decay	1.140310e-07	0.000024	0.000012	0.000016	0.000025	0.000004	0.000002

Table 11: Best hyperparameters of DCAP across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.828248	0.803774	0.639769	0.859296	0.716172	0.722194	0.820315
batch_size	16.000000	16.000000	24.000000	16.000000	24.000000	24.000000	8.000000
dropout	0.167603	0.019843	0.085913	0.038519	0.154270	0.009830	0.000113
epochs	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000	100.000000
lr	0.004886	0.000327	0.001752	0.000259	0.001015	0.000851	0.001752
noise	0.267056	0.080564	0.114802	0.141241	0.013935	0.170603	0.166935
weight_decay	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

Table 12: Best hyperparameters of DeepHit across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.686024	0.713208	0.682997	0.748744	0.714521	0.69207	0.687566
alpha	0.931403	0.598599	0.850615	0.637459	0.90487	0.649244	0.791605
batch_size	16	16	24	16	24	24	8
dropout	0.156714	0.137529	0.164785	0.293165	0.25619	0.153038	0.264258
epochs	200	200	200	200	200	200	200
hidden_layers	[64, 32, 128]	[256, 32, 256]	[32, 32]	[64, 256]	[64, 128]	[128, 256]	[64]
hidden_size_0	64	256	32	64	64	128	64
hidden_size_1	32.0	32.0	32.0	256.0	128.0	256.0	NaN
hidden_size_2	128.0	256.0	NaN	NaN	NaN	NaN	NaN
lr	0.000184	0.000211	0.000349	0.001272	0.002555	0.000129	0.001524
num_layers	3	3	2	2	2	2	1
out_features	20	10	30	10	120	20	120
sigma	0.01186	0.023781	0.086834	0.024053	0.001124	0.063037	0.008641
weight_decay	0.000004	0.0	0.000003	0.000001	0.000002	0.0	0.0

Table 13: Best hyperparameters of DeepSurv across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.676181	0.788679	0.647695	0.48995	0.707921	0.571833	0.641681
batch_size	16	16	24	16	24	24	8
dropout	0.015838	0.219024	0.233182	0.188284	0.190058	0.15858	0.016385
epochs	200	200	200	200	200	200	200
hidden_layers	[32, 64]	[128, 128]	[32, 128, 64]	[32, 256, 64]	[128, 32]	[128, 64]	[128, 128]
hidden_size_0	32	128	32	32	128	128	128
hidden_size_1	64	128	128	256	32	64	128
hidden_size_2	NaN	NaN	64.0	64.0	NaN	NaN	NaN
lr	0.000428	0.000398	0.000107	0.0001	0.004951	0.001731	0.000164
num_layers	2	2	3	3	2	2	2
weight_decay	0.000002	0.000002	0.0	0.000003	0.000003	0.000009	0.00001

Table 14: Best hyperparameters of RSF across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.758366	0.724528	0.583573	0.781407	0.734323	0.678424	0.828021
max_depth	6	7	6	4	7	7	5
min_samples_leaf	25	22	3	11	6	17	16
min_samples_split	14	4	9	3	5	16	9
n_estimators	121	209	119	363	212	569	212

Table 15: Best hyperparameters of Trans-STG across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM_LGG
best_score	0.800197	0.837736	0.706052	0.79397	0.844884	0.717559	0.87986
batch_size	16	16	24	16	24	24	8
d_model	8	16	16	16	8	16	16
dim_feedforward	32	32	8	32	8	8	8
dropout	0.0851292	0.0340481	0.00950718	0.0467148	0.0687669	0.0994098	0.00693402
epochs	200	200	200	200	200	200	200
lr	0.000109483	0.000653411	0.000568071	0.000421422	0.00346371	0.000458326	0.000152026
n_heads	2	1	4	2	8	2	2
n_layers	1	4	1	2	2	3	3
num_time_bins	20	90	120	50	30	30	10
weight_decay	1.06965e-07	8.79884e-06	7.10758e-05	2.09049e-06	3.3555e-07	3.76847e-05	2.57005e-05

Table 16: Best hyperparameters of SurvTRACE across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.807087	0.867925	0.649856	0.836683	0.844884	0.715757	0.860595
attention_dropout	0.0224406	0.0766861	0.0786058	0.0358624	0.0476048	0.0377344	0.0125565
batch_size	16	16	24	16	24	24	8
epochs	200	200	200	200	200	200	200
hidden_dropout	0.0763616	0.0636338	0.10668	0.266337	0.0122195	0.161782	0.0144127
hidden_size	8	16	16	16	16	8	16
intermediate_size	16	8	64	8	32	16	8
lr	0.00176068	0.000270076	0.000174012	0.000669367	0.000380824	0.00227691	0.000894517
num_heads	2	2	2	1	2	2	1
num_durations	50	120	20	20	30	90	50
num_hidden_layers	1	2	1	1	1	2	2
weight_decay	6.44734e-05	8.13339e-07	6.36711e-07	1.26919e-05	9.4674e-07	4.03976e-07	2.2904e-07

Table 17: Best hyperparameters of Transformer Survival across cancers

Hyperparameters	BRCA	CESC	HNSC	SARC	UCEC	COAD _ESCA _READ _STAD	GBM _LGG
best_score	0.754183	0.856604	0.678674	0.806533	0.839934	0.690525	0.744308
T_max	50	30	10	10	30	50	120
alpha	1	1	0.1	0.1	0.5	0.5	1
batch_size	16	16	24	16	24	24	8
d_model	32	128	128	64	64	128	64
dropout	0.21866	0.24803	0.00958298	0.0741388	0.255626	0.0902635	0.0394987
epochs	200	200	200	200	200	200	200
ff_model	256	256	256	128	128	256	512
lr	0.000563374	0.00157532	0.000395509	0.00333269	0.000652544	0.000267775	0.000227006
nhead	8	4	2	2	8	1	8
num_layers	2	4	3	2	1	4	3
weight_decay	4.40966e-06	1.13856e-07	1.96738e-07	6.39507e-06	8.02291e-07	3.10274e-06	7.9478e-07

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